Rabbit Hop

AI shoppers with researched personas shopping real stores on Meta Muse Spark

Track

Indie developer

About me

Akshaya, Executive MBA student, skincare and makeup enthusiast

The origin

Automating customer discovery using agentic infrastructure

The hook

Can an agent shop like a specific kind of person, and can we prove it?

Gemini Notebook

THE PERSONA GAP IN AI AGENTS



TASK COMPLETION

Generic AI agents optimize for speed instead of persona accuracy

Isolated agent runs produce varying and untrustworthy results



PERSONA FIDELITY

Brands need accurate behavioral data before launching new products

Manual discovery surveys capture stated intent rather than real actions

TECHNICAL FRAMING: SUCCESS REQUIRES PERSONA FIDELITY OVER MERE TASK COMPLETION

Gemini Notebook

Pipeline and Architecture



Prior Data

100 synthetic purchase histories mapped to five locked personas

Planning Step

Muse decides search strategy and evidence requirements

Live Shopping

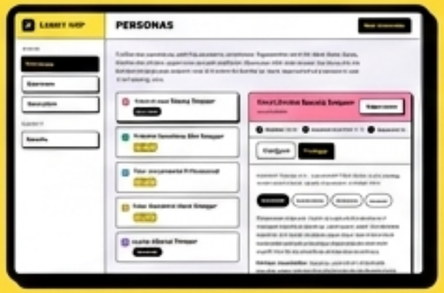
Muse Spark uses computer vision in a real Chrome window

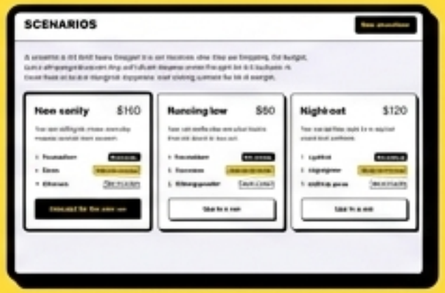
Output Basket

Decisions documented with specific drivers and evidence sources

Evaluation

A blind judge guesses the persona and measures run consistency







Gemini Notebook

ENGINEERING EXECUTION

STACK

- Meta Model API
- Playwright
- Node.js & TypeScript
- React & Vite

PROBLEMS SOLVED



- **Navigation limits:** Bypassed Sephora blocks using new tab forcing and final decision steps
- **Agent shortcuts:** Blocked typed URLs to enforce authentic visual persona planning
- **Meta API constraints:** Handled stateless first-request image rules via computer call outputs
- **Safety checks:** Enforced review policies blocking personal data and sign in pages
- **Reproducibility:** Achieved baseline consistency via zero temperature and versioned prompts

Gemini Notebook

Roadmap and Scaling



Replace archetype histories with authenticated shopper purchase data

Expand retailer support with store specific persona research

Increase automated repeats for statistically meaningful consistency scores

Develop granular persona fidelity scoring metrics per store

Build a fully hosted version requiring direct retail partnerships

Gemini Notebook